

Hurricane Relief Coordination System

Software Requirements Specification

Team Cache Money

Teagan Donnelly, Cameron Seymore, Nicolas Gauvin, Jorge Naranjo, Aden Roof
September 6, 2026 | Version 2.0

Table of Contents

1. Introduction.....	2
2. Stakeholders.....	3
3. Constraints.....	7
4. Overall Description.....	8
5. Business Use Cases.....	9
6. Functional Requirements.....	10
7. Non-Functional Requirements.....	10
8. Competitive Analysis.....	12

1. Introduction

1.1 Purpose

Hurricanes are a dangerous weather event that millions of people experience every year, resulting in the need for aid for thousands of affected people and communities. This project seeks to produce a hurricane relief system application where volunteers may be notified by relief coordinators through the application so that their skills/abilities may be best utilized.

Our goal of this project is to provide a means of effective coordination and task delineation of volunteers to hurricane relief efforts so that responses may be carried out swiftly and effectively. This program seeks to fill the communication gap between the many responding agencies, groups, organizations, and individuals, so that no time or resources are wasted on resolved incidents and so that more people may be helped.

This issue was highlighted in a report by the University of Colorado titled “Relationships Between Affiliated and Unaffiliated Disaster Response Volunteers”. The report was published in 2020 after Hurricane Michael hit Florida and documents volunteer interviews describing showing up to an already handled scene due to lack of communication of its status.

1.2 Scope

This document will cover:

- Potential stakeholders who may have an invested interest in this project
- Any constraints that may apply to the development of this project
- An overall description of the environment the program will operate on
- A competitive analysis to highlight the unique purpose of this project

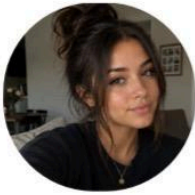
2. Stakeholders

2.1 Stakeholder Overview

Stakeholder Type	Description
Hurricane Survivors	Users needing direct assistance during and after disaster events.
Volunteers	Individuals providing labor, skills, or supplies to help those in need.
Relief Coordinators	Leaders organizing volunteer efforts and allocating resources efficiently.
Disaster Relief Organizations	Large entities managing logistics, supplies, and broader relief strategies.
System Administrators	Personnel responsible for maintaining application infrastructure and ensuring system security.

2.2 Personas

The following personas represent the primary user groups of the Hurricane Relief Coordination System. These personas were developed to illustrate the different needs, goals, resources, and constraints that may affect how users interact with the system.



NAME Sarah Wright

USER TYPE Aid reciever

QUOTE "I'm only 26, but some days I feel like I've already lived a lifetime. I'm tired of being strong because I never really got a choice."

AT A GLANCE

AGE: 26
OCCUPATION: Social Media Influencer
LOCATION: 553 2nd Street East Miami, FL 33134
HOUSEHOLD: Single Mother
TECH COMFORT LEVEL: Comfortable

CURRENT SITUATION

• **ELECTRICITY ACCESS:** No Electricity
 • **INTERNET / CELL SERVICE:** Has Internet
 • **TRANSPORTATION:** Destroyed
 • **HOME ACCESSIBLE?:** No
 • **DISPLACED?:** Yes
 • **DEVICES AVAILABLE:** Smart Phone
 • **OTHER CONSTRAINTS:** Type 2 Diabetes

MOTIVATIONS

• Her child
 • Her work
 • finding emotional support and feeling like herself again
 • Turning her social-media skills into steady income
 • Regaining independence and reliable transportation

GOALS

• Getting out of the hood
 • Professional rapper
 • Social media manager
 • Grow her online audience and find brand partnerships
 • Build a steady income through social media work

NEEDS

• Diabetes medication / insoline
 • Food for her kid
 • Shelter
 • Diapers
 • Transportation

SKILLS / RESOURCES / CAPABILITIES

• Social Media
 • Raising kids
 • Good with phones
 • Physically active

PAIN POINTS / CONSTRAINTS

• Type 2 Diabetes
 • Recent Birth / sea section
 • swollen knees
 • Has a child
 • Post partum depression

TECHNOLOGY & COMMUNICATION

• **PRIMARY DEVICE:** Cell Phone
 • **INTERNET RELIABILITY:** Stable
 • **DIGITAL LITERACY:** Average
 • **ACCESSIBILITY NEEDS:** None
 • **PREFERRED COMMUNICATION:** Call

SCENARIO

After Recently giving birth Hurrican Portia Came and completely Destoryed everything her house is gone her car is gone and Sarah is currently in Post partum. She already suffers from Type 2 diabetes and is currently mentally unstable her emotions are unknown.



NAME Michael Lowe

USER TYPE Volunteer

QUOTE

AT A GLANCE

AGE: 32
 OCCUPATION: Carpenter
 LOCATION: Conway SC
 HOUSEHOLD: Married with kids
 TECH COMFORT LEVEL: Everyday use

CURRENT SITUATION

• ELECTRICITY ACCESS: Yes
 • INTERNET / CELL SERVICE: Internet
 • TRANSPORTATION: Personal Pick-up truck
 • HOME ACCESSIBLE?: Yes
 • DISPLACED?: No
 • DEVICES AVAILABLE: Computer / Phone
 • OTHER CONSTRAINTS:

MOTIVATIONS

- Be a pillar of the community
- Build reputation of his carpenter business
- Be a role model for his kids
- Help the wider area he lives in
-

GOALS

- Have the area he lives in repaired
- Build reputation of his carpenter business
-
-
-

NEEDS

- Way of being contacted for volunteer work
- A way to show certifications
- Easy way of locating who around needs help
- Knowledge of the severity of the area they are going to work in
-

SKILLS / RESOURCES / CAPABILITIES

- Carpentry
- Work bed pick-up truck
- Power tools
- Heavy lifting
-

PAIN POINTS / CONSTRAINTS

- Travel distance
- Sourcing material for aid
- Knowing what equipment will be needed to effectively help
-
-

TECHNOLOGY & COMMUNICATION

• PRIMARY DEVICE: Cell Phone
 • INTERNET RELIABILITY: Stable
 • DIGITAL LITERACY: Average
 • ACCESSIBILITY NEEDS: None
 • PREFERRED COMMUNICATION: Call / Text

SCENARIO

A Hurricane has passed through and Michael wants to help repair homes that may have been damaged but doesn't know the best way to do so

NAME Sarah WilliamsUSER TYPE Relief Coordinator

QUOTE

A goal without a plan is just a wish.

AT A GLANCE

AGE: 42
 OCCUPATION: Emergency Response Director
 LOCATION: Columbia SC
 HOUSEHOLD: Married with kids
 TECH COMFORT LEVEL: Comfortable

CURRENT SITUATION

• ELECTRICITY ACCESS: Yes
 • INTERNET / CELL SERVICE: Yes
 • TRANSPORTATION: Yes
 • HOME ACCESSIBLE?: Yes
 • DISPLACED?: Yes
 • DEVICES AVAILABLE: Yes
 • OTHER CONSTRAINTS: needs info

MOTIVATIONS

- Help people in crisis
- Keep her team safe
- Maintain orderly operations
- Maintain her team's emotional stability during crises.
-

GOALS

- Run her operation efficiently
- Make effective use of her resources.
- Connect volunteers of closest proximity and most applicable skills to nearby people in need.
-

NEEDS

- A method to find people in need of help
- A method to find people willing to volunteer
- A method to evaluate said people's capabilities (ex: owns a truck, can lift heavy loads, etc)
-

SKILLS / RESOURCES / CAPABILITIES

- Organization, managing operations
- leadership under pressure
- Motivating others
- Thinking quickly
-

PAIN POINTS / CONSTRAINTS

- lack of information
- coordinating complex logistics
- Finding people in need
- Managing a large rescue team
-

TECHNOLOGY & COMMUNICATION

• PRIMARY DEVICE: Cell Phone
 • INTERNET RELIABILITY: Stable
 • DIGITAL LITERACY: Average
 • ACCESSIBILITY NEEDS: None
 • PREFERRED COMMUNICATION: Call / Text

SCENARIO

A Hurricane has passed through and Sarah wants to coordinate response efforts. She currently relies on older software as well as calling and texting. She needs a centralized platform which allows her to see people in need on a map, people willing to help, and their respective capabilities, allowing her to organize a more effective response.

3. Constraints

3.1 Time and Schedule

- Formal collaboration is restricted to semi-weekly classroom sessions; additional coordination requires supplemental scheduling.
- Project milestones are subject to strict, predetermined deadlines that must be strictly observed.

3.2 Budgetary Limitations

- The project must operate within a very limited project budget.

3.3 Platform Accessibility

- The solution must ensure full functional accessibility across a diverse range of hardware and software platforms.

3.4 Knowledge and Research

- The project scope is limited by the research and disaster-response knowledge available to the five-person student team.

4. Overall Description

An overview of the environment that this system must operate within, including other software and hardware systems that will interface with this system.

4.1 Operating Environment

The Hurricane Relief System is built to run in high-stress disaster areas where cell service and power are often unreliable. The core system runs on a cloud server to handle backend data, while users interact with the app through mobile devices or web browsers.

4.2 System & Hardware Interfaces

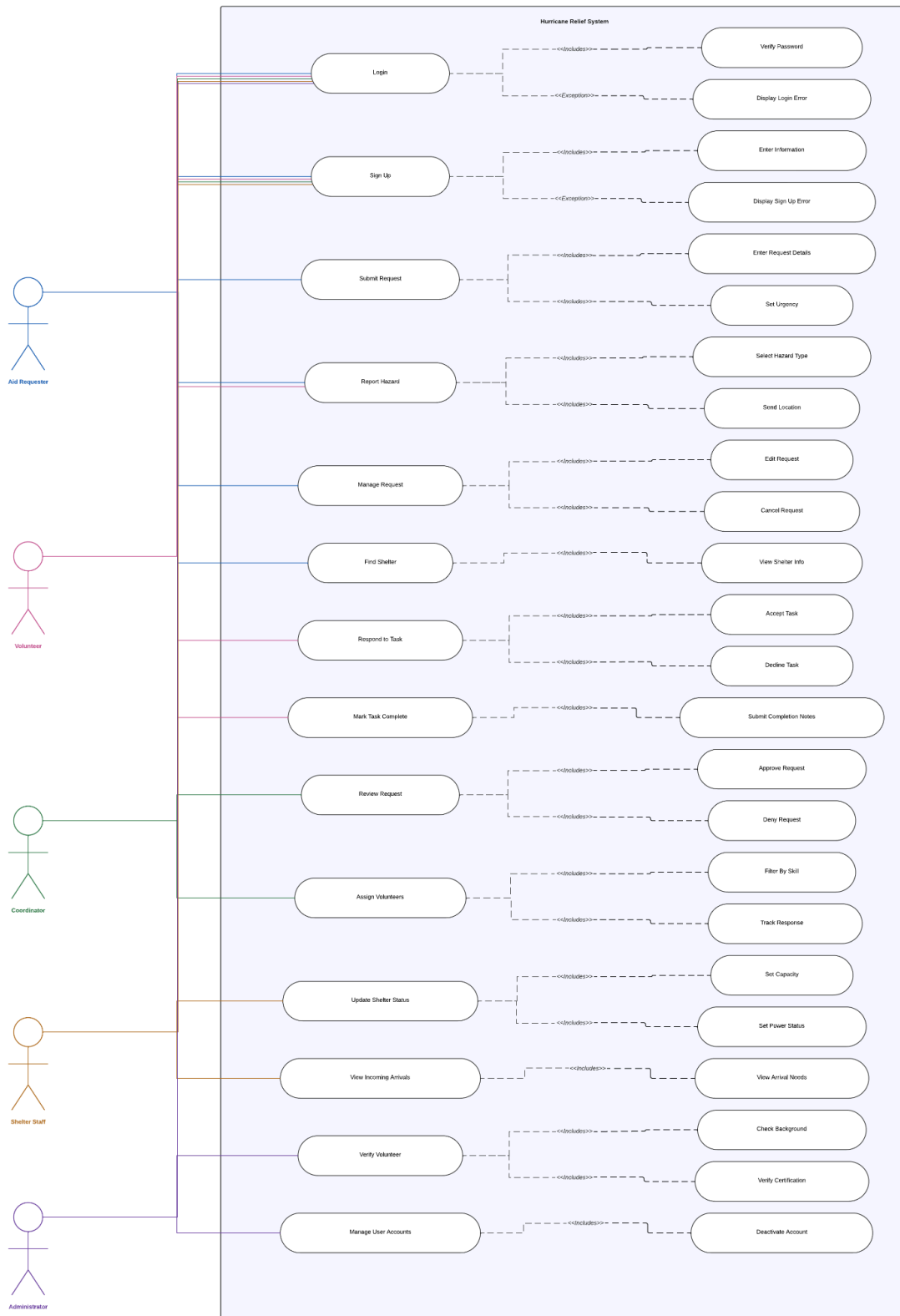
Our program must make use of several existing systems to optimally provide location services and interface with features users are already familiar with:

- **GPS & Location Services:** Uses the phone's built-in GPS to display incident pins on a map and connect the closest volunteers to said locations.
- **Push Notifications & SMS:** Coordinators must be able to send instant alerts and task assignments directly to volunteers' phones, even if internet data is running slow.
- **Emergency Feeds:** Pulls basic public weather and disaster alerts (such as NOAA or FEMA updates) into the coordinator view to help them track conditions in real time.

4.3 System Constraints & Dependencies

- **Connectivity Resilience:** The system must ensure data integrity and operational continuity during periods of intermittent network connectivity or high latency.
- **Data Privacy & Security:** Personally identifiable information (PII) and real-time geospatial data must be encrypted and secured to maintain volunteer and survivor confidentiality.

5. Business Use Cases



6. Functional Requirements

[Requirements Spreadsheet.](#)

7. Non-Functional Requirements

Look and Feel Requirements

- **Clean & Intuitive Interface**
 - Uses high-contrast text and simple visual layouts for easy first-time navigation during emergencies.
 - Features standard color-coded status indicators (e.g., Red for Urgent, Green for Resolved).

Usability Requirements

- **Universal Accessibility**
 - Supports screen readers, high-contrast display modes, and basic physical accessibility standards.
 - Provides options for other languages such as Spanish.

Performance Requirements

- **Low-Bandwidth & Offline Functionality**
 - Saves data offline during power loss and syncs updates within 30 seconds of regaining connectivity.
 - Operates efficiently over slow 3G cellular networks with minimal data usage.

Maintainability and Support Requirements

- **Cross-Platform Compatibility**
 - Supports major web browsers (Chrome, Safari, Firefox) and mobile operating systems (Android, iOS).
 - Built modularly so developers can update features without taking the platform offline.

Security Requirements

- **Data Protection & Secure Access**
 - Encrypts user passwords, personal details, and live location data in transit and at rest.
 - Enforces role-based permissions to hide sensitive victim data from unauthorized users.

Cultural Requirements

- **Community Sensitivity**
 - Uses clear, reassuring language for crisis notifications and automated messages.
 - Tracks dietary needs (Kosher, Halal, Vegan) and household considerations in shelter listings.

Legal Requirements

- **Privacy & Liability Compliance**
 - Complies with local, state, and federal laws regarding personal data privacy.
 - Displays clear liability disclaimers stating the system complements, but does not replace, 911 emergency services.

9. Competitive Analysis

Product	Primary Focus	Key Capabilities	Relevance to Our System
Team Rubicon Roll Call	Disaster volunteer management	Volunteer registration, skill/certification tracking, training, deployment eligibility	Demonstrates the usefulness of structured skill-based volunteer management
American Red Cross Emergency	Emergency information and preparedness	Weather alerts, hurricane tracking, shelter mapping, evacuation guidance, supply checklists	Demonstrates the value of reliable, centralized emergency information
FEMA App	Official disaster information and federal assistance	Disaster alerts, preparedness information, recovery resources, access to federal disaster-assistance information	Demonstrates how survivors can be connected with authoritative recovery resources

9.1 Market Gap Analysis

9.1.1 Market Landscape

The current market is well served by applications focused on weather monitoring and post-disaster damage assessment. Platforms such as Crisis Cleanup and Team Rubicon effectively pinpoint impact zones, allowing for targeted resource allocation. Additionally, real-time alerting remains a standard feature across most established emergency applications.

9.1.2 Identified Gaps

A key gap in existing systems is direct communication between users. Existing solutions primarily provide information in one direction. There is currently no centralized hub or forum-based architecture that allows affected individuals to exchange information directly, seek local assistance, or share real-time updates independently of official channels.

9.1.3 Strategic Opportunities

By integrating robust communication features with the established success of damage-pinpointing technology, there is a strategic opportunity to move beyond passive information delivery. The proposed solution will address the "information-only" limitation of current apps by providing a centralized communication platform that enables users to coordinate community-level relief efforts directly.